Hassan Imani

hasanimani.pa@gmail.com | LinkedIn: hassan-imani-parashkooh | +1 (778) 513-1039

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | AML AI Research & Special Projects — Jul 2025 – Present
Scotiabank — *Toronto, ON*

- **Behavioral Anomaly Detection:** Refactored and optimized the team's outlier detection model conditioned on KYC profiles, isolating peer-group transactional anomalies to suppress false positives.
- **Representation Learning & Pipelines:** Learned self-supervised client representations from temporal transaction sequences, establishing validation suites measuring semantic alignment against cohort descriptors.
- **ML Platform & Synthetic Benchmarking:** Engineered an Hyperparameter Optimization tool and built synthetic transactional data generators to benchmark unsupervised pipelines.
- **Production Alert Pipelines:** Implemented end-to-end detection pipelines from distributed SQL/feature extraction through model inference to LLM-enabled alert narratives.

Machine Learning Engineer — Feb 2024 – Jul 2025
Zown — *Toronto, ON*

- **Listing Matching Engine:** Owned end-to-end design and deployment of a multi-stage candidate matching engine; paired hard-filtering with a neural Two-Tower architecture and auxiliary multi-task MLPs predicting dwell time and engagement.
- **Continuous Training & Valuation Systems:** Deployed deep tabular property valuation models backed by an automated Continuous Training (CT) pipeline and holdout validation gating to guard against production metric degradation.
- **Visual Representation Modeling:** Fine-tuned a Vision Transformers (ViT) with multi-task classification and regression heads to extract granular listing features directly into downstream ranking systems.
- **Cold-Start Optimization:** Implemented dynamic exploration-set strategies to resolve user cold-start and ensure candidate catalog diversity.

EDUCATION

York University — 2021 – 2025
PhD in Mechanical Engineering — *Toronto, ON*

- Equivariant Graph Neural Networks (eGNNs), spatio-temporal trajectory modeling, and statistical regression for high-dimensional dynamical systems.

Tehran Polytechnic — 2018 - 2021
MSc in Mechanical Engineering — *Tehran, IR*

Sharif University of Technology — 2013 - 2018
BSc in Mechanical Engineering — *Tehran, IR*

TECHNICAL SKILLS

Languages & Distributed: Python, SQL, Bash, PySpark, Spark

ML & Statistics: Anomaly Detection, Representation Learning, Recommender Systems, Graph Neural Networks (GNNs), Vision Transformers

Tools & Platforms: Git, Docker, GCP, MongoDB, PyTorch, Copilot

PUBLICATIONS

Graph Neural Networks for Sustainable Energy (HI Parashkooh, C Jian) — 2025
Predicting Adsorption in Aromatic Molecules — *ChemE 9 (4), 85*

An Innovative Method for Calculating PMV (HI Parashkooh, C Jian) — 2024
from a standard molecular trajectory — *J. Mol. Liq. 395, 123879*

Data Mining Guided Molecular Investigations (HI Parashkooh, C Jian) — 2022
on the Coalescence of Water-in-Oil Droplets — *Energy Fuels 36 (4), 1811*